



Graphs in Machine Learning

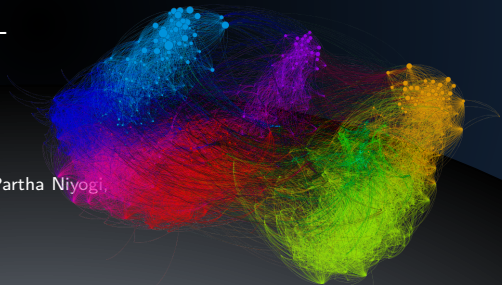
Transductive Generalization Bounds

Stability-Based Bounds for SSL

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Partially based on material by: Mikhail Belkin, Partha Niyogi,
Olivier Chapelle, Bernhard Schölkopf



Transductive Generalization Bounds

True risk vs. empirical risk

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$$R_P(f) = \frac{1}{N} \sum_i (f_i - y_i)^2$$
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We look for **transductive** bounds in the form

$$R_P(f) \leq \hat{R}_P(f) + \text{errors}$$

Transductive Generalization Bounds

Bounding transductive error using stability analysis

<http://www.cs.nyu.edu/~mohri/pub/str.pdf>

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$$\ell^* = \min_{\ell \in \mathbb{R}^N} (\ell - \mathbf{y})^T \mathbf{C}(\ell - \mathbf{y}) + \ell^T \mathbf{Q} \ell$$

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Closed form solution

$$\ell^* = (\mathbf{C}^{-1} \mathbf{Q} + \mathbf{I})^{-1} \mathbf{y}$$

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By the generalization bound of Belkin **belkin2004regularization**,
w.p.1 $-\delta$

http://web.cse.ohio-state.edu/~mbelkin/papers/RSS_COLT_04.pdf

$$R_P(\ell^*) \leq \underbrace{\widehat{R}_P(\ell^*) + \beta + \sqrt{\frac{2 \ln(2/\delta)}{n_I}}(n_I \beta + 4)}_{\text{transductive error } \Delta_T(\beta, n_I, \delta)} .$$

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Bounding **transductive** error

$$\|\ell_2^* - \ell_1^*\|_\infty \leq \beta \leq \frac{\|\mathbf{y}_2 - \mathbf{y}_1\|_2}{\frac{\lambda_m(\mathbf{Q})}{\lambda_M(\mathbf{C}_1)} + 1} + \frac{\lambda_M(\mathbf{Q})\|\mathbf{C}_1^{-1} - \mathbf{C}_2^{-1}\|_2 \cdot \|\mathbf{y}_1\|_2}{\left(\frac{\lambda_m(\mathbf{Q})}{\lambda_M(\mathbf{C}_2)} + 1\right) \left(\frac{\lambda_m(\mathbf{Q})}{\lambda_M(\mathbf{C}_1)} + 1\right)}$$

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Now, let us plug in the values for our problem.

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Take $c_l = 1$

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This algorithm is β -stable!

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We have an idea how to set γ_g !

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<https://misovalko.github.io/mva-ml-graphs.html>